Ideation Phase
Define the Problem Statements

Date	8-04-2026
Team ID	SWTID-2026-2088
Project Name	Automated Data Import and Relationship mapping using input sets & Dot walking.
Maximum Marks	2 Marks

Customer Problem Statement :

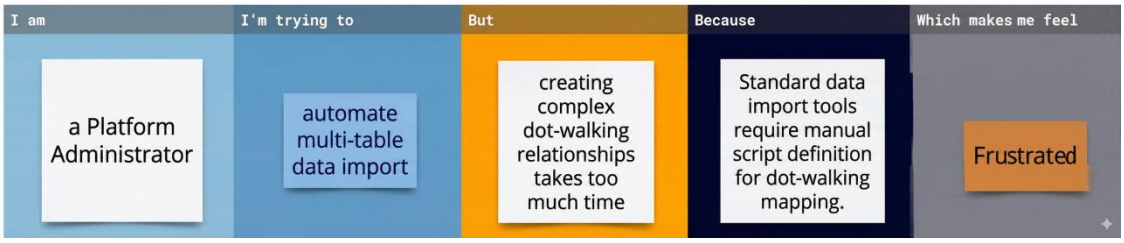
In modern enterprise systems, large volumes of data are frequently imported from multiple external sources such as spreadsheets, databases, and third-party applications. These datasets often contain complex relationships between entities, which must be accurately established within the target system. However, existing data import processes are largely manual or semi-automated, making them time-consuming, error-prone, and difficult to scale.

Additionally, identifying and mapping relationships between different data entities requires significant effort, especially when dealing with hierarchical or relational data structures. The lack of an efficient mechanism to dynamically traverse and link related records further complicates the process.

This project aims to address these challenges by developing an automated solution that leverages structured input sets for data ingestion and utilizes dot walking techniques to intelligently map relationships between entities. The system will reduce manual intervention, improve data accuracy, and streamline the overall data import and relationship mapping process.

I am	Describe customer with 3-4 key characteristics - *who are they?*	A Platform Administrator responsible for onboarding new data sources and managing complex entity relationships within a large ecosystem.
I'm trying to	*List their outcome or "job" the care about - what are they trying to achieve?*	Import multi-level data sets and automatically create correct, deep relationship mappings between distinct tables and entities (e.g., connecting a 'Task' to an 'Incident' to a 'User' in one go).
but	Describe what problems or barriers stand in the way – *what bothers them most?*	The traditional manual mapping methods are extremely slow and error-prone, especially when defining deep, multi-hop relationships. Creating and maintaining scripts to handle complex dot-walking across multiple tables is difficult and requires extensive platform knowledge.
because	Enter the "root cause" of why the problem or barrier exists – *what needs to be solved?*	Establishing relationships across tables that do not have a direct foreign key is complex. The platform supports "dot-walking" (traversing through related tables), but configuring these relationships is typically a manual, brittle process, and standard import sets don't handle this automatically out of the box.
which makes me feel	Describe the emotions from the customer's point of view – *how does it impact them emotionally?*	Frustrated by the time-consuming configuration, anxious about introducing data errors and broken links during the import process, and overwhelmed by the complexity of maintaining custom integration logic for every new source.

Reference: https://www.tableau.com/learn/articles/guide-to-data-mapping?utm_source=chatgpt.com



Problem Statement (PS)	**I am (Customer)**	**I'm trying to**	**But**	**Because**	**Which makes me feel**
PS-1	A ServiceNow Platform Developer.	Automatically create parent-child	Creating the `Dot Walking` paths for these mappings *manually*	The standard `Input Set` transformation engine doesn't support recursive	Overwhelmed by the constant need for maintenance and anxious that my custom scripts will introduce critical data-integrity errors.
PS-2	An HR Data Steward responsible for importing weekly employee data from a third-party application.	Use the new automated import system to quickly map employee .	I can't do this *myself* within the new system's user interface	The current implementation of automated mapping with dot-walking is still a "black box"	Impatient, dependent, and frustrated that a simple data management task takes an enterprise-level IT ticket to resolve.